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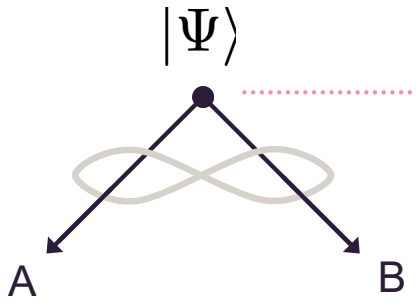
Week 13, Lecture 2
Entanglement

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03

Entanglement

Entanglement in quantum technology



entangled pair is generated

each qubit in the entangled pair can be sent to spatially separated parties

Entanglement is useful in:

- Quantum teleportation
- Quantum key distribution
- Quantum computing

Quantum entanglement

Example 1

$|\psi\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |11\rangle)$ can be written as $|\psi\rangle = \frac{1}{\sqrt{2}}|1\rangle \otimes (|0\rangle + |1\rangle)$ } Separable not entangled

Example 2

$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ cannot be written as $|\psi_1\rangle \otimes |\psi_2\rangle$ } Not separable Entangled

Definition: A bipartite pure quantum state $|\psi\rangle \in \mathcal{H}^{12}$ is seaparble if it can be written as $|\psi\rangle = |\psi_1\rangle \otimes |\psi_1\rangle$. A quantum state is entangled if it is not separable.

Bell states

The Bell states are maximally entangled states of two qubits

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$
$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$
$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$
$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Measuring one qubit in the computational basis would reveal that the second qubit is in the same state (for $|\Phi^\pm\rangle$) or opposite state (for ($|\Psi^\pm\rangle$)

$\{|\Phi^+\rangle, |\Phi^-\rangle, |\Psi^+\rangle, |\Psi^-\rangle\}$ form an orthonormal basis spanning the Hilbert space of a two-qubit system, known as the Bell basis

Pure entangled states of two qubits

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$
$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$
$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$
$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Bell states
Pure and maximally entangled

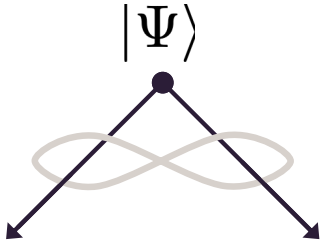
Other examples of **pure** two qubit states:

$$|\psi\rangle = \sqrt{\frac{3}{5}}|00\rangle + \sqrt{\frac{2}{5}}|11\rangle$$
$$|\psi\rangle = \sqrt{\frac{2}{5}}(|00\rangle + |11\rangle) + \sqrt{\frac{1}{5}}|01\rangle$$

Not maximally entangled
But still entangled

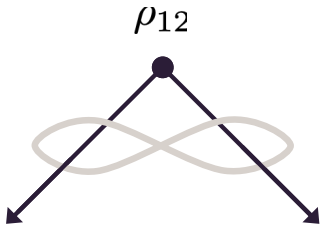
Mixed entangled states of two qubits

pure states



Definition: A bipartite pure quantum state $|\psi\rangle \in \mathcal{H}^{12}$ is separable if it can be written as $|\psi\rangle = |\psi_1\rangle \otimes |\psi_1\rangle$

mixed states

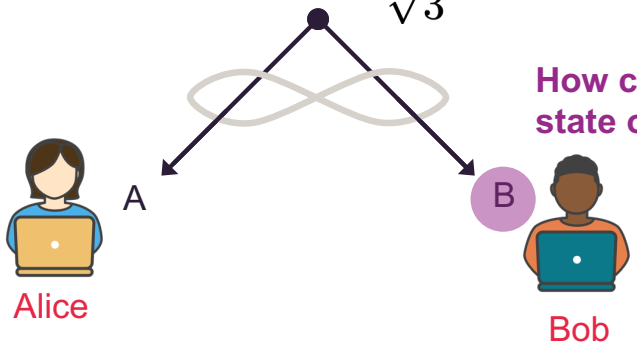


Definition: A bipartite mixed quantum state $\rho \in \mathcal{H}^{12}$ is separable if it can be written as

$$\rho = \sum_i p_i \rho_1^i \otimes \rho_2^i$$

Reduced density matrix

$$|\psi\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |11\rangle + |01\rangle)$$



How can we describe the state of the system at B?

$$|\psi\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |11\rangle + |01\rangle)$$
$$= \sqrt{\frac{2}{3}}|0\rangle_A \frac{1}{\sqrt{2}}(|0\rangle_B + |1\rangle_B) + \sqrt{\frac{1}{3}}|1\rangle_A |1\rangle_B$$

If alice measures her qubit:

- $|0\rangle_A$ with probability $\frac{2}{3}$ → the state of bob's qubit must be: $\frac{1}{\sqrt{2}}|0\rangle_B + |1\rangle_B$
- $|1\rangle_A$ with probability $\frac{1}{3}$ → $|1\rangle_B$

Density matrix at B:

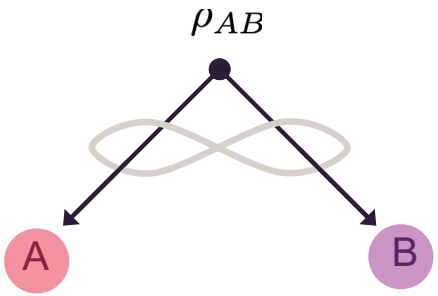
$$\rho_B = \frac{2}{3}\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\frac{1}{\sqrt{2}}(\langle 0| + \langle 1|) + \frac{1}{3}|1\rangle\langle 1|$$
$$= \frac{2}{3}\begin{bmatrix}\frac{1}{2} & \frac{1}{2}\\ \frac{1}{2} & \frac{1}{2}\end{bmatrix} + \frac{1}{3}\begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}$$
$$= \begin{bmatrix}\frac{1}{3} & \frac{1}{3}\\ \frac{1}{3} & \frac{2}{3}\end{bmatrix}$$

Reduced density matrix of subsystem B

Partial trace

$$\rho_A = \sum_{n_B} \langle n_B | \rho_{AB} | n_B \rangle = \mathrm{Tr}_B [\rho_{AB}]$$

Where $|n_B\rangle$ are the basis vectors for subsystem B

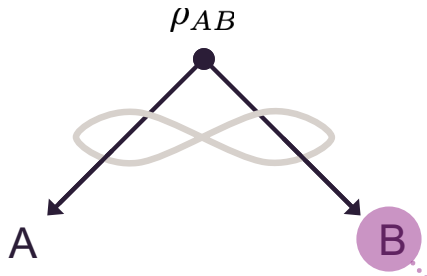


$$\rho_B = \sum_{n_A} \langle n_A | \rho_{AB} | n_A \rangle = \mathrm{Tr}_A [\rho_{AB}]$$

Where $|n_A\rangle$ are the basis vectors for subsystem A

- The operation for obtaining the reduced density matrices of a composite quantum system is called the partial trace.
- To obtain the density matrix of subsystem A, we perform the partial trace over subsystem B
- To obtain the density matrix of subsystem B, we perform the partial trace over subsystem A
- The reduced density matrix for an entangled pure ensemble is a mixed ensemble.

Partial trace - example



How can we describe the state of the system at B?

$$|\psi\rangle = \frac{1}{\sqrt{3}}\left(|00\rangle + |11\rangle + |01\rangle\right)$$

$$\rho_{AB} = \frac{1}{3}\left(|00\rangle + |11\rangle + |01\rangle\right)\left(\langle 00| + \langle 11| + \langle 01|\right)$$

$$= \frac{1}{3}\left(|00\rangle\langle 00| + |00\rangle\langle 11| + |00\rangle\langle 01| + |11\rangle\langle 00| + |11\rangle\langle 11| + |11\rangle\langle 01| + |01\rangle\langle 00| + |01\rangle\langle 11| + |01\rangle\langle 01|\right)$$

$$\rho_B = \mathrm{Tr}_A\left[\rho_{AB}\right]$$

$$= \sum_{n_A} \langle n_A|\,\rho_{AB}\,|n_A\rangle$$

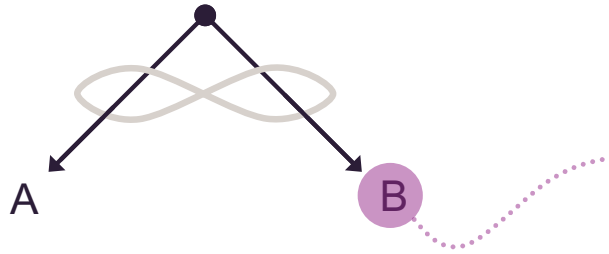
$$= \langle 0|_A\,\rho_{AB}\,|0\rangle_A + \langle 1|_A\,\rho_{AB}\,|1\rangle_A$$

$$= \frac{1}{3}\left(|0\rangle\langle 0| + |0\rangle\langle 1| + |1\rangle\langle 0| + |1\rangle\langle 1|\right) + \frac{1}{3}|1\rangle\langle 1|$$

$$= \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

Partial trace

$$|\psi\rangle = \frac{1}{\sqrt{3}}\left(|00\rangle + |11\rangle + |01\rangle\right)$$



How can we describe the state of the system at B?

$$\rho_{AB} = \begin{array}{c} \begin{matrix} \langle 0|_A \langle 0|_B & \langle 0|_A \langle 1|_B & \langle 1|_A \langle 0|_B & \langle 1|_A \langle 1|_B \end{matrix} \\ \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \end{bmatrix} \end{array} \begin{matrix} |0\rangle_A |0\rangle_B \\ |0\rangle_A |1\rangle_B \\ |1\rangle_A |0\rangle_B \\ |1\rangle_A |1\rangle_B \end{matrix}$$

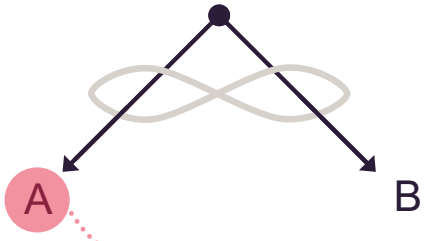
$$\rho_B = \sum_{n_A} \langle n_A| \rho_{AB} |n_A\rangle = \mathrm{Tr}_A\left[\rho_{AB}\right]$$

$$\rho_B = \frac{1}{3}\left(|0\rangle_B \langle 0|_B + |0\rangle_B \langle 1|_B + |1\rangle_B \langle 0|_B + |1\rangle_B \langle 1|_B + |1\rangle_B \langle 1|_B\right)$$

$$\rho_B = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

Partial trace

$$|\psi\rangle = \frac{1}{\sqrt{3}}\left(|00\rangle + |11\rangle + |01\rangle\right)$$



How can we describe the state of the system at **A**?

$$\rho_{AB} = \begin{array}{c} \begin{matrix} \langle 0|_A \langle 0|_B & \langle 0|_A \langle 1|_B & \langle 1|_A \langle 0|_B & \langle 1|_A \langle 1|_B \end{matrix} \\ \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \end{bmatrix} \end{array} \begin{matrix} |0\rangle_A |0\rangle_B \\ |0\rangle_A |1\rangle_B \\ |1\rangle_A |0\rangle_B \\ |1\rangle_A |1\rangle_B \end{matrix}$$

$$\rho_A = \sum_{n_B} \langle n_B | \rho_{AB} | n_B \rangle = \mathrm{Tr}_B \left[\rho_{AB}\right]$$

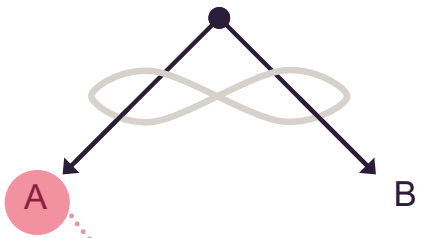
$$\rho_A = \frac{1}{3}\left(|0\rangle_A \langle 0|_A + |0\rangle_A \langle 0|_A + |0\rangle_A \langle 1|_A + |1\rangle_A \langle 0|_A + |1\rangle_A \langle 1|_A\right)$$

$$\rho_A = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

Partial trace – another example

Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)$$



$$\rho_{AB} = \frac{1}{2}\left(|00\rangle + |11\rangle\right)\left(\langle 00| + \langle 11|\right)$$

$$= \begin{bmatrix} \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{bmatrix}$$

How can we describe the state of the system at **A**?

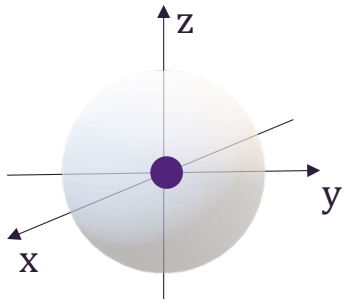
$$\rho_A = \mathrm{Tr}_B\left[\rho_{AB}\right]$$

$$= \sum_{n_B} \langle n_B|\, \rho_{AB}\, |n_B\rangle$$

$$= \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$$

this state is maximally mixed

on the bloch sphere



Classical entropy

The entropy of a discrete random variable X is:

$$H(X) := -\sum_x p_x \log_2 p_x$$

also called the Shannon entropy

where p_x is the probability of obtaining result x and the sum is over all possible values of x.

In **classical information theory**, entropy measures **uncertainty** or **average information content** in a random variable.

Entropy quantifies *our ignorance* about the outcome before observing it.

Quantum entropy

In quantum mechanics, we also have the **von Neumann entropy**:

$$S(\rho) = -\mathrm{Tr}\,(\rho \log_2 \rho)$$

ρ is the density operator describing the quantum state

$S(\rho) = 0$: system is in a pure state

$S(\rho) > 0$: system is in mixed state

- **Spectral decomposition**

$$\hat{\rho} = \sum_i \lambda_i \left|\phi_i\right\rangle \left\langle\phi_i\right| \quad \Rightarrow \quad S = -\mathrm{Tr}\left[\hat{\rho} \ln \hat{\rho}\right] = -\sum_i \lambda_i \ln \lambda_i$$

eigenvalues

eigenvectors

Joint entropy

The **joint entropy** measures the **total uncertainty** of two random variables considered together.

Classical

$$H(X,Y) := -\sum_{x,y} p_{x,y} \log_2 p_{x,y}$$

Quantum

$$S(\rho_{AB}) = -\mathrm{Tr}\left(\rho_{AB} \log_2 \rho_{AB}\right)$$

Joint entropy is greater than or equal to the maximum of the individual entropies:

$$H(X,Y) \geq \max\left[H(X), H(Y)\right]$$

different for quantum

Joint entropy

Classical

In classical physics, if two bits (A and B) are perfectly correlated — say:

$$(0,0) \qquad \text{or} \qquad (1,1)$$

each with 50% probability:

$$P(0,0) = P(1,1) = \frac{1}{2}$$

$$H(X,Y) := -\sum_{x,y} p_{x,y} \log_2 p_{x,y} = 1$$

joint system also has nonzero entropy

Marginal entropies

$$H(X) = H(Y) = 1$$

each subsystem appears *mixed* (uncertain)

no pure, deterministic global state — only classical correlations between random variables.

Quantum

For the quantum Bell state:

$$|\Psi\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Global system is pure

$$S(\rho_{AB}) = 0$$

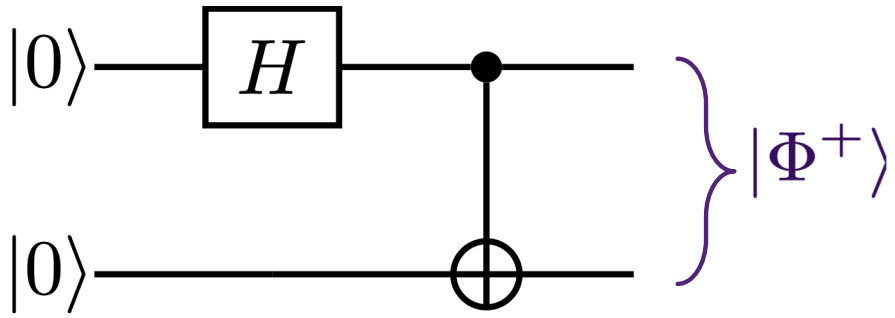
Subsystems are mixed

$$S(\rho_A) = S(\rho_B) = 1$$

in the **quantum case**, the total entropy is **zero**, even though each part looks maximally random

entropy of the whole is less than the sum of its parts, reflecting **nonclassical correlations (entanglement)**.

Bell state generation



this circuit takes the four two-qubit states: → to the four bell states:

$$|00\rangle \dashrightarrow |\Phi^+\rangle$$
$$|01\rangle \dashrightarrow |\Psi^+\rangle$$
$$|10\rangle \dashrightarrow |\Phi^-\rangle$$
$$|11\rangle \dashrightarrow |\Psi^-\rangle$$

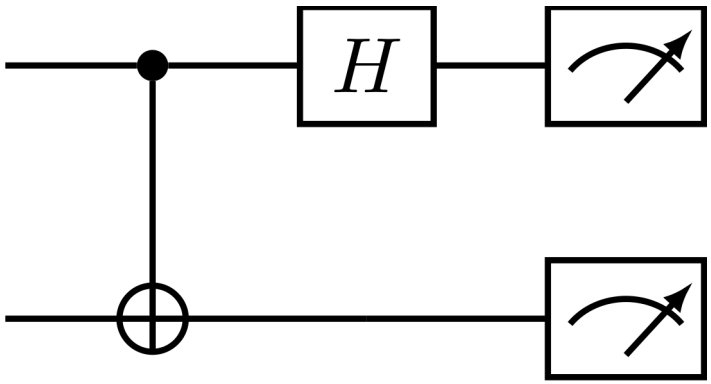
The first part of this circuit is a Hadamard gate on the first qubit. In matrix representation:

$$H \otimes I = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

The second part of this circuit is a CNOT gate between the first and second qubits

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Measurements in the Bell basis



measure in the computational basis

Input the four bell states into this circuit

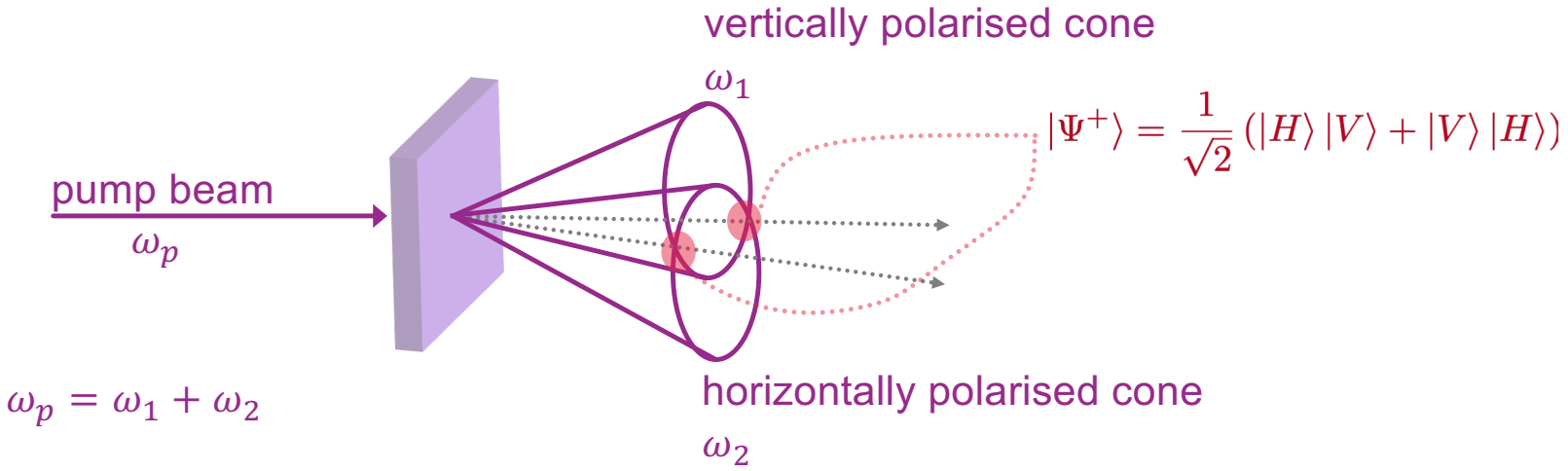
The computational basis measurement results at the end will be:

$$|\Phi^+\rangle \dashrightarrow |00\rangle$$
$$|\Psi^+\rangle \dashrightarrow |01\rangle$$
$$|\Phi^-\rangle \dashrightarrow |10\rangle$$
$$|\Psi^-\rangle \dashrightarrow |11\rangle$$

- This circuit is the inverse of the circuit shown on the previous slide.
- It performs Bell state decoding: we can determine which of the four Bell states the two qubits input into this circuit are in.
- The Bell measurement corresponds to measuring the joint observables:

$$Z \otimes Z \text{ and } X \otimes X$$

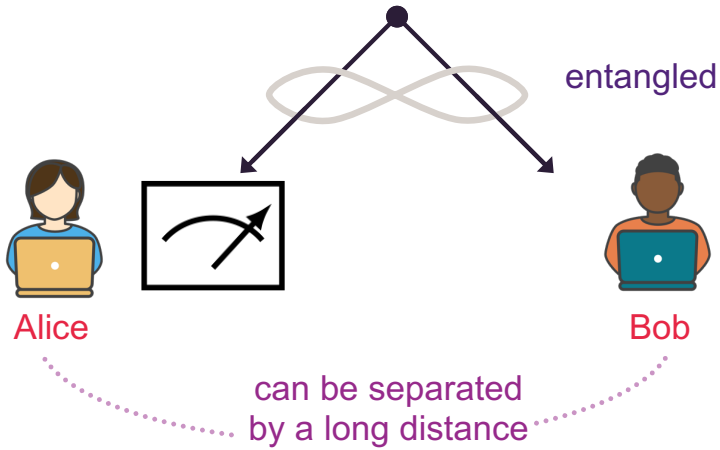
Entangled photon pair generation



- Parametric down-conversion generates pairs of photons that can be entangled in various degrees of freedom.
- A single photon from a pump laser at angular frequency ω_p is incident on a birefringent crystal and down-coverted into a pair of photons at angular freqencies ω_1 and ω_2, with $\omega_p = \omega_1 + \omega_2$ (energy entanglement).
- In certain configurations, the down-converted photons are emitted in cones with orthogonal polarisation. At the intersection points of the cones, the light is in a polarisation entangled Bell state.

Entanglement and local hidden variables

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$



Can these correlations be explained by a local hidden variable?

Local hidden variable theories ask whether there may be underlying unknown hidden variables that are causing the correlations in measurement outcomes

measurement outcome

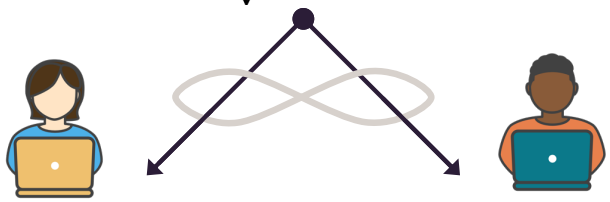
If Alice measures $|0\rangle$ then Bob's qubit is in $|0\rangle$ state

If Alice measures $|1\rangle$ then Bob's qubit is in $|1\rangle$state

} correlated measurement outcomes

Bell-CHSH inequality

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle_A |1\rangle_B - |1\rangle_A |0\rangle_B\right)$$



Alice

Bob

Alice's measurements

$$A_0 = \sigma_z$$

$$A_1 = \sigma_x$$

Bob's measurements

$$B_0 = -\frac{1}{\sqrt{2}}(\sigma_x + \sigma_z)$$

$$B_1 = \frac{1}{\sqrt{2}}(\sigma_x - \sigma_z)$$

expectation values:

$$\langle A_0 B_0\rangle = \langle A_0 B_1\rangle = \langle A_1 B_0\rangle = \frac{1}{\sqrt{2}} \qquad \langle A_1 B_1\rangle = -\frac{1}{\sqrt{2}}$$

$$\langle A_0 B_0\rangle + \langle A_0 B_1\rangle + \langle A_1 B_0\rangle - \langle A_1 B_1\rangle = 2\sqrt{2}$$

greater than the value of 2 predicted by classical local hidden variables

Bell-CHSH inequality:

$$\langle A_0 B_0\rangle + \langle A_0 B_1\rangle + \langle A_1 B_0\rangle - \langle A_1 B_1\rangle \leq 2$$

- The Bell inequality is a used to check that quantum entanglement cannot be caused by local hidden variables.
- The inequality gives a mathematical constraint on the correlations of the outcomes of two measurements performed independently.
- Quantum physics predicts that this inequality is violated.

Peres-Horodecki criterion (PPT criterion)

For two qubits:

$$\rho_{AB} = \begin{bmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{bmatrix}$$

each A_{nm} is a 2x2 matrix

The partial transpose of ρ_{AB} with respect to subsystem B is:

$$\rho^{T_B} = (I \otimes T)\,\rho_{AB}$$

identity on A

transpose on B

$$\rho^{T_B} = \begin{bmatrix} A_{00}^T & A_{01}^T \\ A_{10}^T & A_{11}^T \end{bmatrix}$$

each A_{nm} is transposed

Peres-Horodecki (PPT) criterion:
If ρ_{AB} is separable, then all eigenvalues of ρ^{T_B} are non-negative

- In general, the problem of determining whether quantum systems are entangled or not is NP-hard in the system size, however, for some specific cases there are efficient criteria for separability
- The Peres-Horodecki criterion, or positive partial transpose (PPT) criterion is a way of verifying whether a bipartite quantum state is entangled or separable.
- You can tranpose either subsystem because $\rho^{T_A} = (\rho^{T_B})^T$

PPT criterion - example

Given the following two qubit state:

$$\rho_w = \frac{4F-1}{3}\left|\Phi^+\right\rangle\left\langle\Phi^+\right| + \frac{1-F}{3}\mathrm{I}_4$$

Use the PPT criterion to determine for what value of F the state is entangled.

$$\rho_w = \frac{4F-1}{3}\frac{1}{2}\left(|00\rangle + |11\rangle\right)\left(\langle 00| + |11\rangle\right) + \frac{1-F}{3}\mathrm{I}_4$$

$$= \begin{bmatrix} \frac{1+2F}{6} & 0 & 0 & \frac{4F-1}{6} \\ 0 & \frac{1-F}{3} & 0 & 0 \\ 0 & 0 & \frac{1-F}{3} & 0 \\ \frac{4F-1}{6} & 0 & 0 & \frac{1+2F}{6} \end{bmatrix}$$

The partial transpose with respect to subsystem B:

$$\rho_w^{T_B} = \begin{bmatrix} \frac{1+2F}{6} & 0 & 0 & 0 \\ 0 & \frac{1-F}{3} & \frac{4F-1}{6} & 0 \\ 0 & \frac{4F-1}{6} & \frac{1-F}{3} & 0 \\ 0 & 0 & 0 & \frac{1+2F}{6} \end{bmatrix}$$

The eigenvalues of $\rho_w^{T_B}$:

$$\lambda_1 = \underbrace{\frac{1-2F}{2}} \qquad \lambda_2 = \lambda_3 = \lambda_4 = \frac{1+2F}{6}$$

negative for $F > 0.5$ → ρ_w is entangled for $F > 0.5$

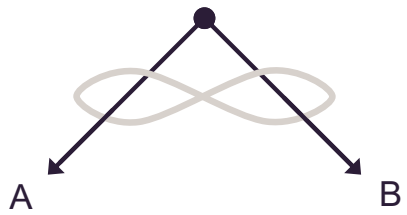
Entangled higher-dimensional systems

A single qudit:

$$|\psi\rangle = a_0 |0\rangle + a_1 |1\rangle + a_2 |2\rangle + \cdots + a_{d-1} |d-1\rangle$$ dim:d

Maximally entangled qudit state:

$$|\psi\rangle = \frac{1}{\sqrt{d}} \sum_{n=0}^{d-1} |n\rangle |n\rangle$$



If we were to trace out one mode:

$$\rho_A = \sum_{n_B} \langle n_B | \rho_{AB} | n_B \rangle = \mathrm{Tr}_B [\rho_{AB}]$$
$$\rho_B = \sum_{n_A} \langle n_A | \rho_{AB} | n_A \rangle = \mathrm{Tr}_A [\rho_{AB}]$$

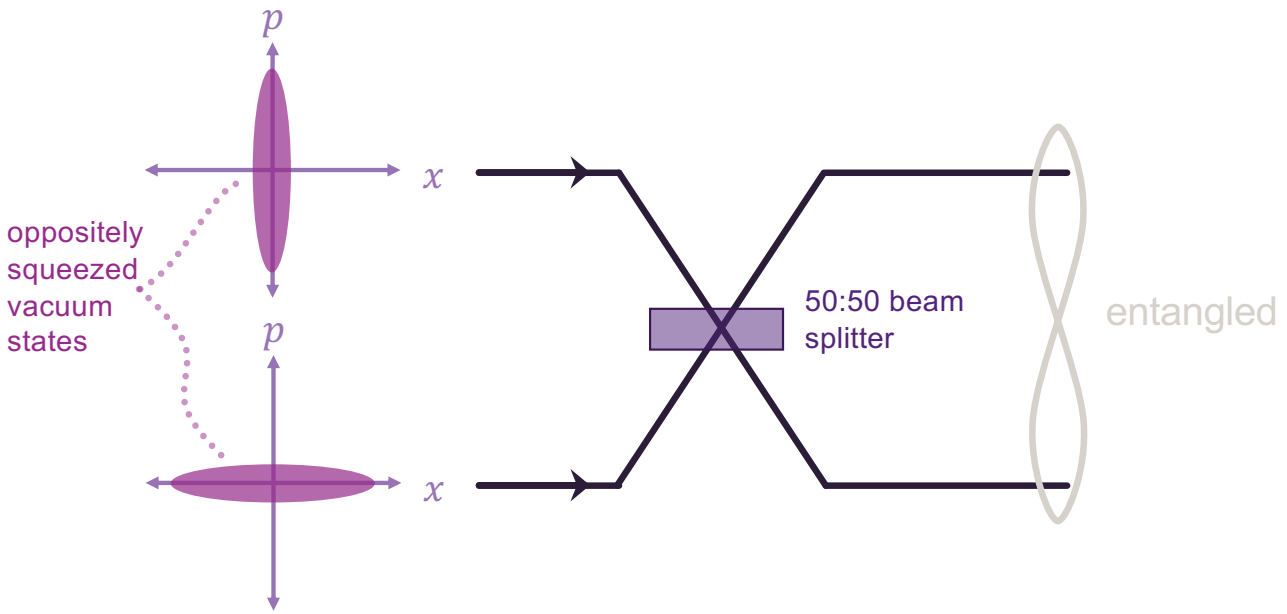
} these reduced density matrices would be maximally mixed

Two-mode squeezed vacuum states

- If we combine two squeezed vacuum states, with opposite squeezing parameters, on a 50:50 beam splitter, the output will be the entangled two-mode squeezed vacuum state (TMSV)
- This state has quadrature correlations as well as photon number correlations

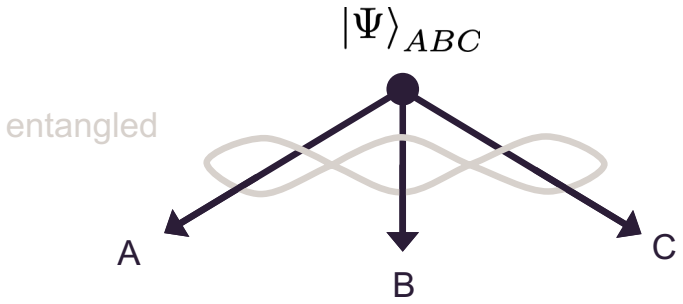
$$|TMSV\rangle = \sqrt{1-\lambda^2}\sum_{n}^{\infty}\lambda^n |n\rangle |n\rangle$$

$$\lambda = \tanh r$$



- This state is maximally entangled only in the limit as $\lambda \to \infty$ (but this limit is not physically achievable as it requires infinite energy)

Tripartite entanglement



$$|GHZ\rangle = \frac{1}{\sqrt{2}}\left(|000\rangle + |111\rangle\right)$$

$$|W\rangle = \frac{1}{\sqrt{3}}\left(|001\rangle + |010\rangle + |100\rangle\right)$$

not biseparable
genuine three qubit entanglement

- The GHZ and W states are examples of genuine tripartite entangled states of three qubits – they are both not biseparable
- The GHZ and W states can both be extended to more than three qubits. For n-qubits, these states are

$$|W\rangle = \frac{1}{\sqrt{n}}\left(\underbrace{|100\cdots0\rangle}_{\text{n-qubits}} + |010\cdots0\rangle + \cdots|000\cdots1\rangle\right)$$

$$|GHZ\rangle = \frac{1}{\sqrt{2}}\left(\overbrace{|000\cdots\rangle}^{\text{n-qubits}} + |111\cdots1\rangle\right)$$

Summary

What we talked about today…

01

Quantum information

02

Photons as qubits

03

Entanglement